2D ResCNN Neural Demapper Architecture & Theory

In-Depth Technical Report: Mathematical Foundations, Tensor Flow, and Layer Design

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1. CORE THEORY: WHY A 2D VISION ARCHITECTURE FOR OFDM?

Exploiting Time-Frequency Coherence Surfaces Without Pilot Symbols

1.1 The Physical Nature of OFDM Resource Grids

In modern multi-carrier communications (5G NR), transmission is structured over a 2D time-frequency matrix comprising N_sym = 14 OFDM symbols along the time axis and N_sc = 64 subcarriers along the frequency axis, yielding 896 Resource Elements (REs). The wireless propagation channel imposes two continuous physical correlations across this grid:

• **Frequency Coherence (Coherence Bandwidth B_c):** Neighboring subcarriers experience closely correlated fading gains because their frequency separation (15 kHz) is smaller than the channel coherence bandwidth. If subcarrier k experiences a phase shift and attenuation, subcarrier k+1 undergoes nearly identical perturbation.

• **Time Coherence (Coherence Time T_c):** Within a single 1 ms 5G slot, Doppler shifts are negligible for moderate vehicular speeds, causing consecutive OFDM symbols to share identical or smoothly evolving channel coefficients.

1.2 How the 2D CNN Replaces Classical LS/LMMSE

Classical receivers treat each subcarrier in isolation after noisy Least-Squares (LS) interpolation. In contrast, our **2D Residual CNN** slides 2D convolutional kernels across the entire 14x64 surface. The receptive field captures multi-subcarrier and multi-symbol correlation patterns, allowing the neural network to **implicitly estimate the channel, denoise fading distortions, and compute soft bit probabilities simultaneously** without dedicated pilot reference tones.

2. INPUT & OUTPUT TENSOR SPECIFICATIONS

Complete Mathematical Dimensions and Bit-Plane Mapping

Tensor Stage	Mathematical Shape	Physical Meaning & Structure Description
Complex Grid	Y in C^[B, 14, 64]	Raw received frequency-domain baseband matrix after FFT at receiver.
Input Tensor	Y_tilde in R^[B, 2, 14, 64]	Channel 0 = Real (In-phase I), Channel 1 = Imaginary (Quadrature Q).
Hidden Features	F_l in R^[B, 48, 14, 64]	48 learned non-linear spatial feature maps extracted across 3 ResBlocks.
Output Logits	Lambda_grid in R^[B, 4, 14, 64]	4 soft LLR planes corresponding to the 4 bit planes of 16-QAM.
Flattened LLRs	lambda_hat in R^[B, 3584]	Permuted [B, 14, 64, 4] -> Reshaped to 3584 soft LLRs for 5G LDPC BP decoding.

Understanding Soft Log-Likelihood Ratios (LLRs):

The output vector contains real-valued soft log-odds for each coded bit: **LLR(c_i) = ln[P(c_i = 1 | Y) / P(c_i = 0 | Y)]**. Positive values indicate confident '0' bits, negative values indicate confident '1' bits, and values near zero represent high channel uncertainty.

3. LAYER-BY-LAYER ARCHITECTURAL SPECIFICATIONS

Detailed Structural Breakdown of the ~90,000 Parameter Backbone

Layer / Block	Input Dim	Output Dim	Mathematical Formula & Structural Implementation
Input Projection	[B, 2, 14, 64]	[B, 48, 14, 64]	F0 = GELU(GroupNorm_1(Conv2D(2->48, k=3x3, p=1, bias=True)))
ResBlock 1	[B, 48, 14, 64]	[B, 48, 14, 64]	F1 = GELU(GN(Conv(GELU(GN(Conv(F0))))) + F0) [Skip Highway]
ResBlock 2	[B, 48, 14, 64]	[B, 48, 14, 64]	F2 = GELU(GN(Conv(GELU(GN(Conv(F1))))) + F1) [Skip Highway]
ResBlock 3	[B, 48, 14, 64]	[B, 48, 14, 64]	F3 = GELU(GN(Conv(GELU(GN(Conv(F2))))) + F2) [Skip Highway]
Output Conv Head	[B, 48, 14, 64]	[B, 4, 14, 64]	Z = Conv2D(48->4, k=3x3, p=1); Calibrated LLRs = alpha * Z
Permute & Reshape	[B, 4, 14, 64]	[B, 3584]	permute(0, 2, 3, 1).contiguous().view(B, -1)

4. MATHEMATICAL JUSTIFICATION OF DESIGN CHOICES

4.1 Group Normalization (G = 1) vs. Batch Normalization

Standard BatchNorm relies on batch statistics (mean, variance) calculated across mini-batches. During real-time receiver operation, frames arrive individually (batch size B = 1), causing BatchNorm statistics to destabilize. **GroupNorm (with G = 1)** computes normalization over all 48 channels and the 14x64 grid for each frame independently: **mu = (1 / CHW) sum x**, **sigma^2 = (1 / CHW) sum (x - mu)^2**. This guarantees deterministic behavior during both single-frame streaming and batched training.

4.2 GELU (Gaussian Error Linear Unit) Activation

Standard ReLU hard-clips all negative inputs to zero (causing dead neurons when channel noise fluctuates). **GELU: GELU(x) = x * Phi(x) = x * P(X <= x)** provides smooth, continuous, non-zero curvature for small negative inputs, allowing faint channel gradients in deep fading notches to propagate back without vanishing.

4.3 Identity Residual Skip Connections

The skip connection **F_l = GELU(F_{l-1} + Residual(F_{l-1}))** creates a direct mathematical gradient highway: **dLoss / dF_{l-1} = (dLoss / dF_l) * (dResidual / dF_{l-1} + I)**. The identity matrix I prevents gradient attenuation, allowing the network to learn subtle perturbation corrections on top of preserved baseband features.

4.4 Learnable Temperature Scaling (alpha) for LLR Calibration

Downstream 5G LDPC Belief Propagation decoders pass soft messages between variable and check nodes. Overconfident LLRs saturate check nodes (tanh(x) -> 1.0), destroying soft information, while underconfident LLRs slow convergence. Our learnable scalar **alpha in R^+** is optimized via backpropagation to dynamically scale LLR logits to match the exact noise variance N_0.